

Selectime experiment guideline

1 Question and scope

Test whether validation can select useful Chronos-2 input choices without parameter training. Preserve Adaptime’s 90 dataset/frequency/term tasks and its four dataset exclusions. Compare univariate, multivariate, past-only square-root-magnitude/sign augmentation, top- K for $K = 1, 5, 10, 15$, and both task-level and item/variant-level validation-selected forecasts. Accuracy benefits are hypotheses, not obtained evidence.

2 Protocol

The official TIME test length and horizon H define the unchanged test windows. The default validation length is $H + (\lfloor T_{\text{test}}/H \rfloor - 1)s_{\text{val}}$, with cadence s_{val} inherited from Adaptime and no reserved training interval. Truncate validation at the series beginning if necessary. Model contexts use observed history up to 8192 steps; retrieval uses its configured fixed lookback. Apply the inherited task-specific alignment period and datastore stride, and require complete maximum- K support. Datastore caps are optional and disabled by default.

Validation retrieval ends strictly at validation start; test retrieval expands to the pre-test interval after selection. Neighbor futures are finite, wholly before the applicable boundary, and available before the query. Use instance-normalized Euclidean lookback distance and rescale retrieved pasts and futures to query units with past-only statistics. All-series scope is default; exact same-series scope remains an explicit ablation.

3 Selection and randomness

Select all candidate methods by per-date MSSE and the paired bootstrap one-standard-error rule. Use 1000 replications by default. The automatic block length is the larger of rounded $N^{1/3}$ and $\lceil H/s_{\text{val}} \rceil$, capped at $N - 1$. Prefer univariate, multivariate, self-augmentation, then smaller K within the uncertainty threshold. With no usable dates select univariate; with one date use the observed minimum. One run seed (default zero) controls backbone stochasticity and bootstrap resampling. No component seed exists.

4 Execution and artifacts

Hydra configuration is `src/timebench/conf/experiment.yaml`. Dataset settings are `src/timebench/config/datasets.yaml`. The main entry point is `src/scripts/run_selectime.py`. The standalone selectors are `select_per_task.py` and `select_per_variate.py` in the same directory. They require completed validation forecasts. Scheduler fronts source the common `src/slurm/` workflow, which runs preparation, validation retrieval and forecasting, both selectors,

expanded test retrieval and forecasting, selected-prediction assembly, evaluation, and report in order.

Use an already prepared saved-Arrow dataset and local Chronos-2 checkpoint. The selected Seasonal Naive grid is a prerequisite for test forecasting. All project products live under **outputs/selectime/**; logs under **logs/**. Every stage allocates a schema-1 configuration-addressed manifest before work. Reuse exact completed runs and restart interrupted tasks from their beginning. Reports retain consumed manifests, metric coverage, population dispersion, fallbacks and validation selection counts. Selected-method independent inference latency remains unmeasured. Forecast stages finalize canonical vanilla after its successful scheduler step before forecasting the other candidates. Report configuration policies are **error/distinct/latest/average**; repeat policies are **selected/latest/distinct/average**. The default uses the current exact configuration and selected repeat. Explicit filters and distinct labels retain nested producer settings. Average repeat task statistics within each exact configuration before averaging configurations; retain every consumed manifest.

5 Verification and evidence

Lightweight scientific regression source exists under **src/tests/**. Synthetic and static regression checks cover chronological datastore boundaries, aligned retrieval, normalization, query-scale covariates, bootstrap selection, the exact 90-task configuration, scheduler fronts and report policies. They do not exercise the foundation backbone or a cluster environment. Obtain a narrow remote smoke result before the full study, then analyze complete matched artifacts. No results are currently available.